Caching and stale information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete cache eligibility record.

Keep query, trusted access context, versions, save/current times, TTL, decision and evidence together.

Keep one habit: Recall → W1–W3/W5 → independent TTL/version case → cache project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries.

- **Cache key** includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.

Data retention and deletion

> **My remembered idea, example and correction**

Untrusted input boundaries

> **My remembered idea, example and correction**

Approval gates and idempotency

My remembered idea, example and correction

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A cache stores a previous result for possible reuse. Check current trusted permission before serving a value. The classroom key is query, trusted role, corpus version and procedure version. Use virtual minutes: age = now minus saved time; valid age satisfies 0 ≤ age < TTL, so equality expires.

  Main TTL 5: a teacher c1/p1 entry saved at 0 hits at 4 only with permission; a student same-query request misses, revoked permission denies, and age 5 expires. Corpus c2 or procedure p2 misses even at age 1. Separate picture: approved version 5 says 09:30, while saved version 4 says 09:00.

  Hold role/procedure fixed; first version 5 request misses, an authorized local fixture lookup supplies 09:30, and a later matching fresh request may hit. Silent unversioned source edits cannot be detected by the same key and TTL. Roles and permissions are supplied fixtures, not a login system; finer access differences require additional key scope.

  Freshness does not establish truth, and expiry does not prove retention deletion. The code uses copied fictional dictionaries and virtual times only.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A cache stores a previous result for possible reuse. Check current trusted permission before serving a value. The classroom key is query, trusted role, corpus version and procedure version. Use virtual minutes: age = now minus saved time; valid age satisfies 0 ≤ age < TTL, so equality expires.

  Main TTL 5: a teacher c1/p1 entry saved at 0 hits at 4 only with permission; a student same-query request misses, revoked permission denies, and age 5 expires. Corpus c2 or procedure p2 misses even at age 1. Separate picture: approved version 5 says 09:30, while saved version 4 says 09:00.

  Hold role/procedure fixed; first version 5 request misses, an authorized local fixture lookup supplies 09:30, and a later matching fresh request may hit. Silent unversioned source edits cannot be detected by the same key and TTL. Roles and permissions are supplied fixtures, not a login system; finer access differences require additional key scope.

  Freshness does not establish truth, and expiry does not prove retention deletion. The code uses copied fictional dictionaries and virtual times only.

- Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A cache stores a previous result for possible reuse. Check current trusted permission before serving a value. The classroom key is query, trusted role, corpus version and procedure version. Use virtual minutes: age = now minus saved time; valid age satisfies 0 ≤ age < TTL, so equality expires.

  Main TTL 5: a teacher c1/p1 entry saved at 0 hits at 4 only with permission; a student same-query request misses, revoked permission denies, and age 5 expires. Corpus c2 or procedure p2 misses even at age 1. Separate picture: approved version 5 says 09:30, while saved version 4 says 09:00.

  Hold role/procedure fixed; first version 5 request misses, an authorized local fixture lookup supplies 09:30, and a later matching fresh request may hit. Silent unversioned source edits cannot be detected by the same key and TTL. Roles and permissions are supplied fixtures, not a login system; finer access differences require additional key scope.

  Freshness does not establish truth, and expiry does not prove retention deletion. The code uses copied fictional dictionaries and virtual times only.

- A report omits this qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A cache stores a previous result for possible reuse. Check current trusted permission before serving a value. The classroom key is query, trusted role, corpus version and procedure version. Use virtual minutes: age = now minus saved time; valid age satisfies $0 \le age < TTL$, so equality expires.

  Main TTL 5: a teacher c1/p1 entry saved at 0 hits at 4 only with permission; a student same-query request misses, revoked permission denies, and age 5 expires. Corpus c2 or procedure p2 misses even at age 1. Separate picture: approved version 5 says 09:30, while saved version 4 says 09:00.

  Hold role/procedure fixed; first version 5 request misses, an authorized local fixture lookup supplies 09:30, and a later matching fresh request may hit. Silent unversioned source edits cannot be detected by the same key and TTL. Roles and permissions are supplied fixtures, not a login system; finer access differences require additional key scope.

  Freshness does not establish truth, and expiry does not prove retention deletion. The code uses copied fictional dictionaries and virtual times only.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- TTL 10 minutes; ages 9, 10, 11. Corpus version changes at age 9. Decide validity.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled caching and stale information **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of caching and stale information.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Caching and stale information. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Caching and stale information. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- TTL 10 minutes; ages 9, 10, 11. Corpus version changes at age 9. Decide validity.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

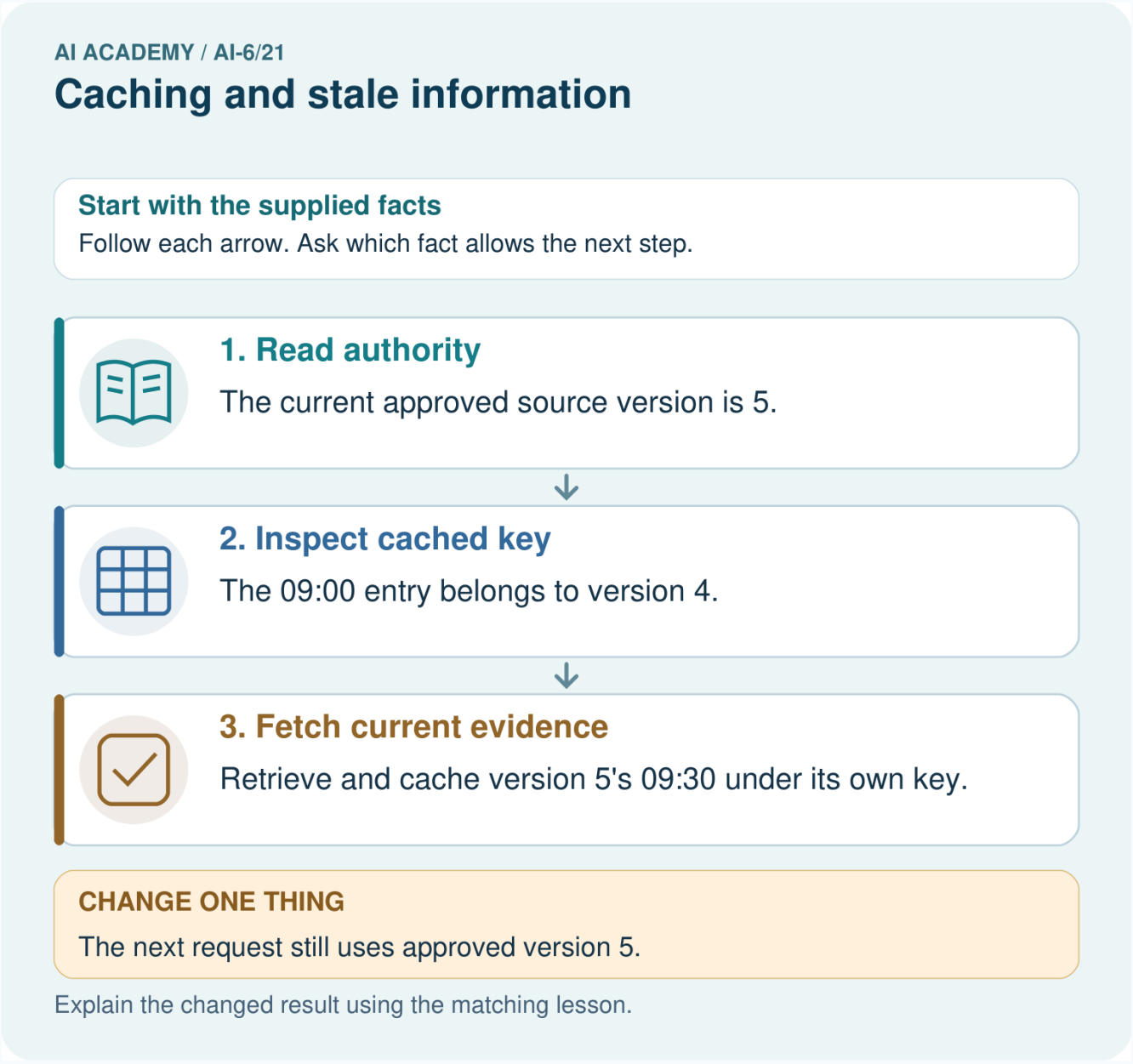
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.
- Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.
- Identify the version mismatch in the cached 09:00 answer. What should the first version 5 request return, and what permits a later version 5 cache hit?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.
- Which labelled step matches this detail? Retrieve and cache version 5's 09:30 under its own key.
- 1. Read authority
- 2. Fetch current **evidence**

- 3. Inspect cached key

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

<details>
<summary>Check after the first attempt</summary>

After your attempt, compare with the supported explanation: Fetch current **evidence**
Retrieve and cache version 5's 09:30 under its own key. A cache saves repeated work but can retain obsolete facts.

Version-aware keys distinguish answers grounded in different records.
</details>

Changed-case question and optional hint

- Changed-case practice: The next request still uses approved version 5. Which conclusion is supported by the supplied facts?

- 1. The matching version 5 entry can supply 09:30 under the freshness policy.

- 2. This one example guarantees correct results for every future **input**.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The matching version 5 entry can supply 09:30 under the freshness policy. Request and cached source version now agree.

Repair and retain the record

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Approval gates and idempotency (ai-6/15). Goal: Explain the **mechanism** and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9.

How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

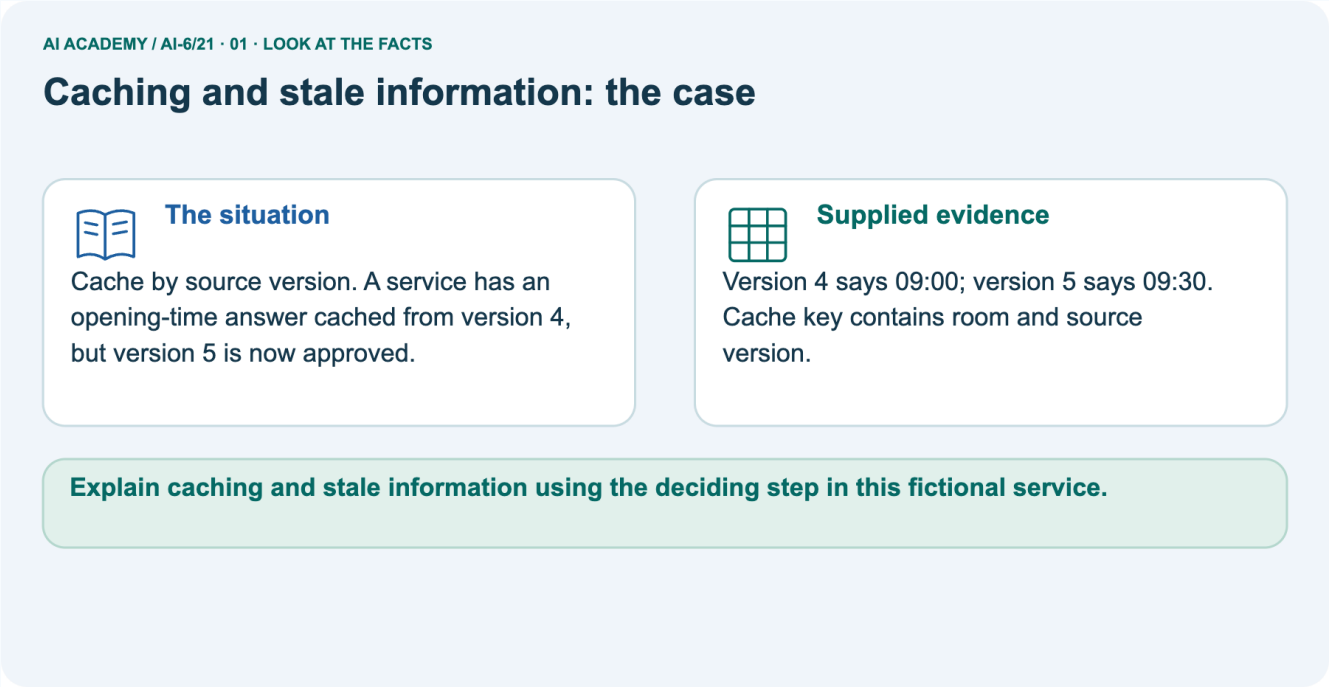
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Caching and stale information: the case



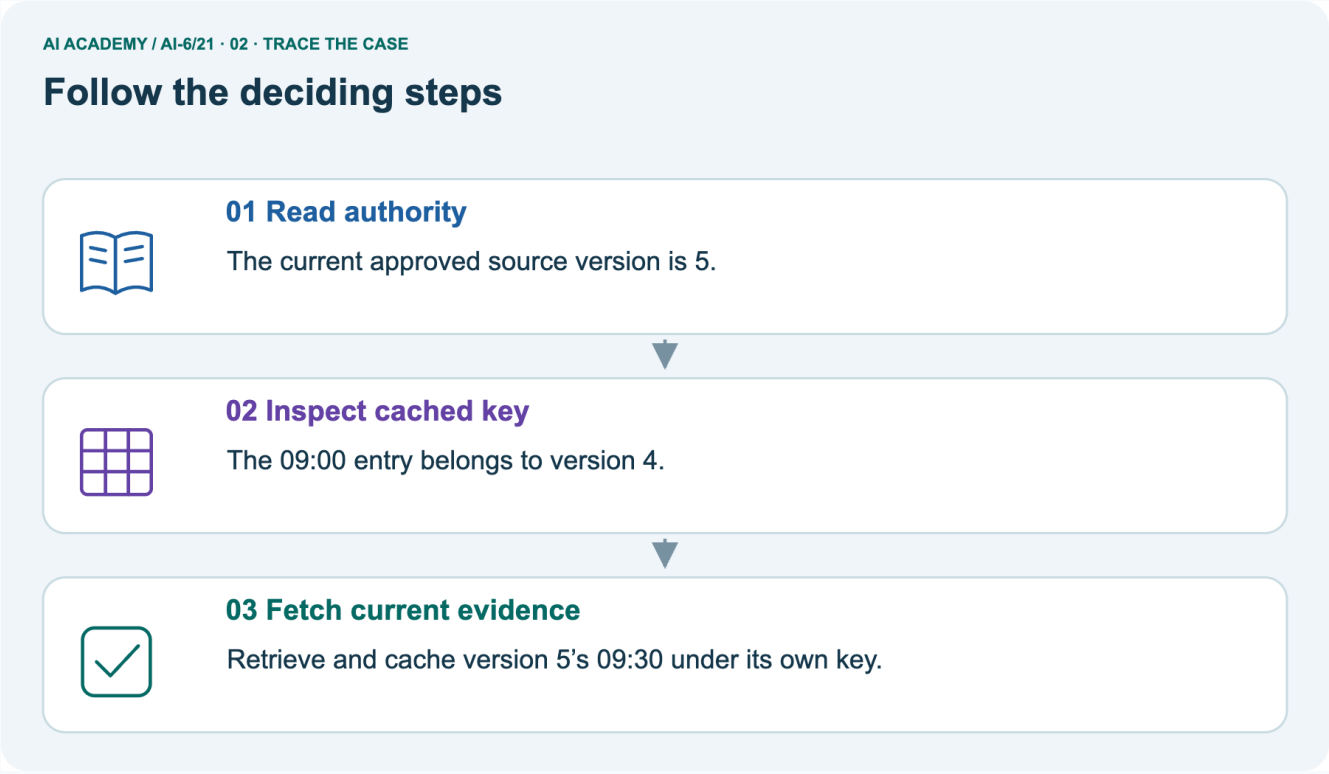
Follow the labelled picture with the matching explanation. Open the vector diagram

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved. Supplied facts: Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



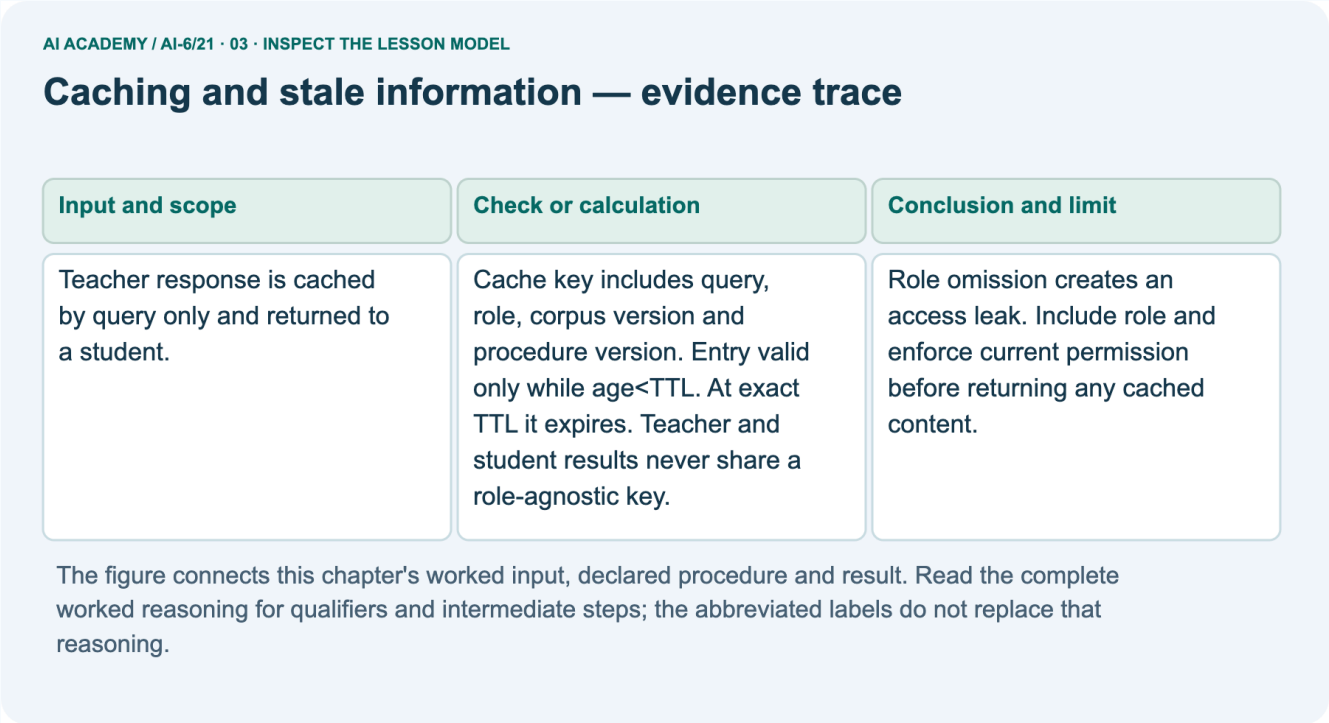
Follow the labelled picture with the matching explanation. Open the vector diagram

Read authority: The current approved source version is 5. → Inspect cached key: The 09:00 entry belongs to version 4. → Fetch current **evidence**: Retrieve and cache version 5's 09:30 under its own key.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Caching and stale information — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

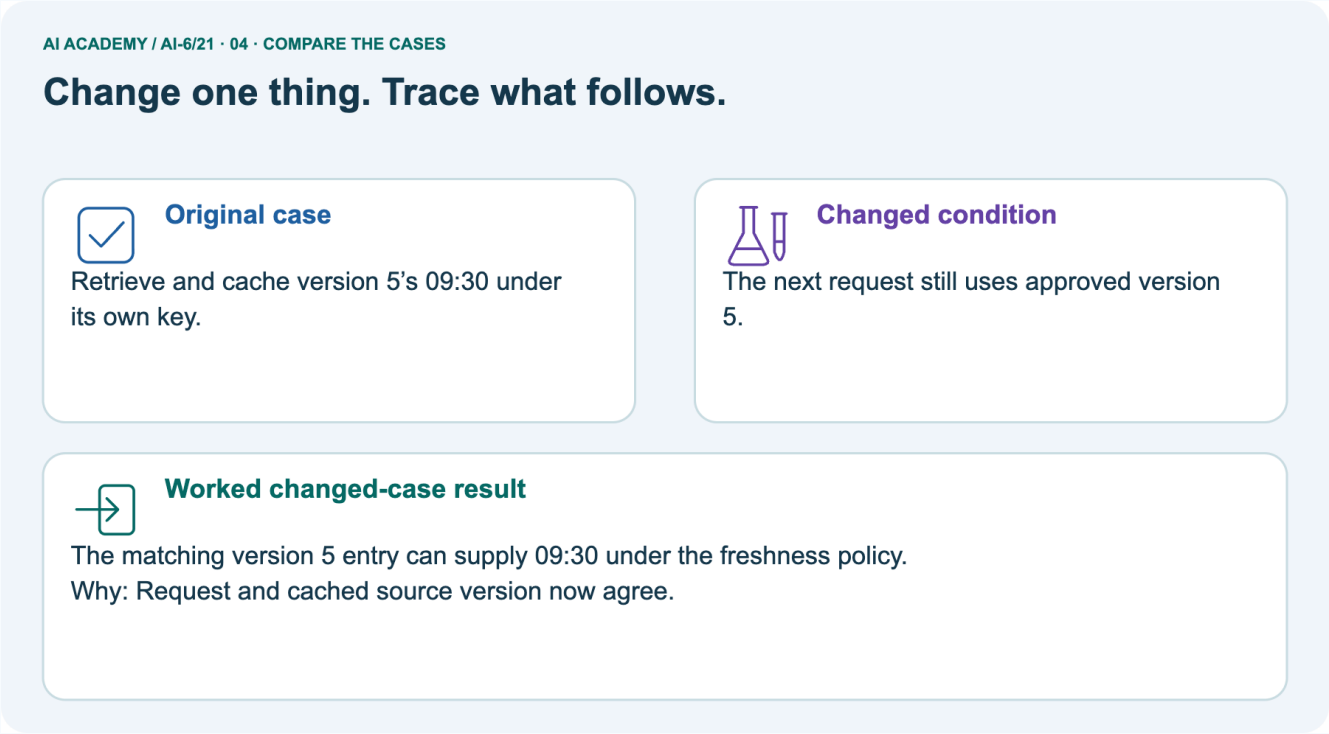
Caching and stale information — **evidence** trace. Teacher response is cached by query only and returned to a student. | **Cache key** includes query, role, corpus version and procedure version. Entry valid only while age<TTL.

At exact TTL it expires. Teacher and student results never share a role-agnostic key. | Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



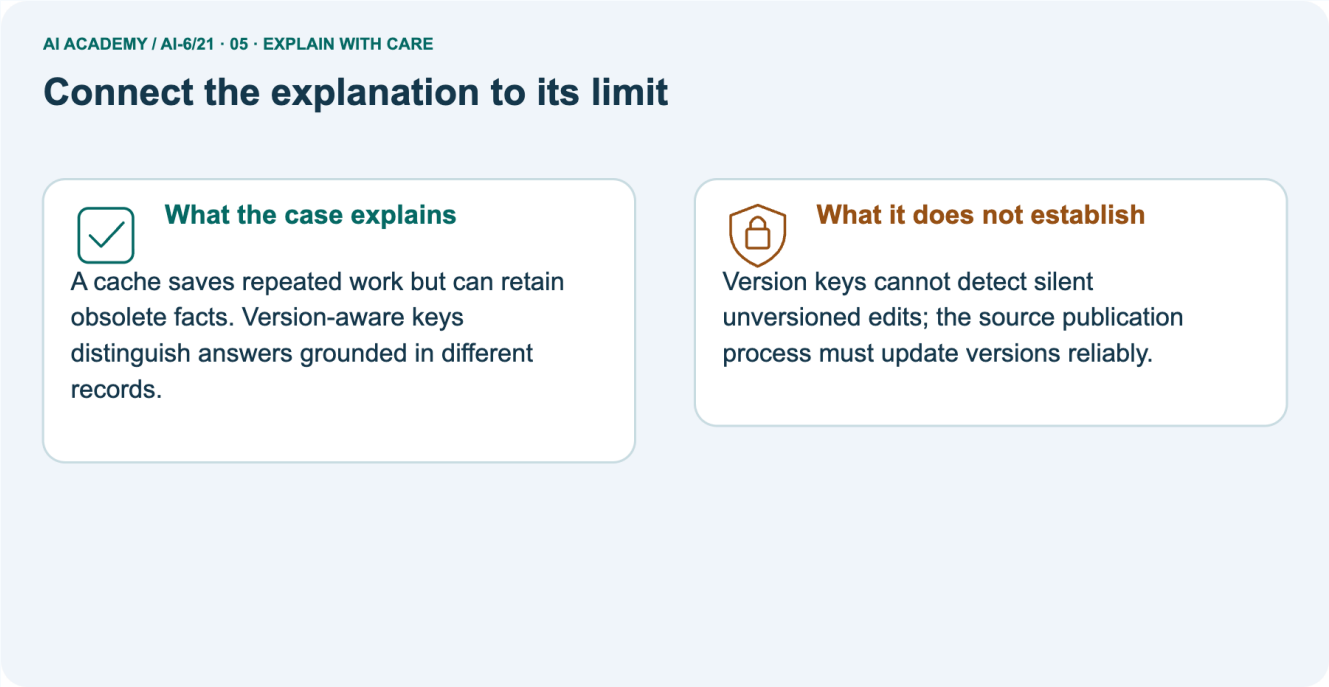
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retrieve and cache version 5's 09:30 under its own key. Changed condition: The next request still uses approved version 5. Expected result: The matching version 5 entry can supply 09:30 under the freshness policy. Reason: Request and cached source version now agree.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

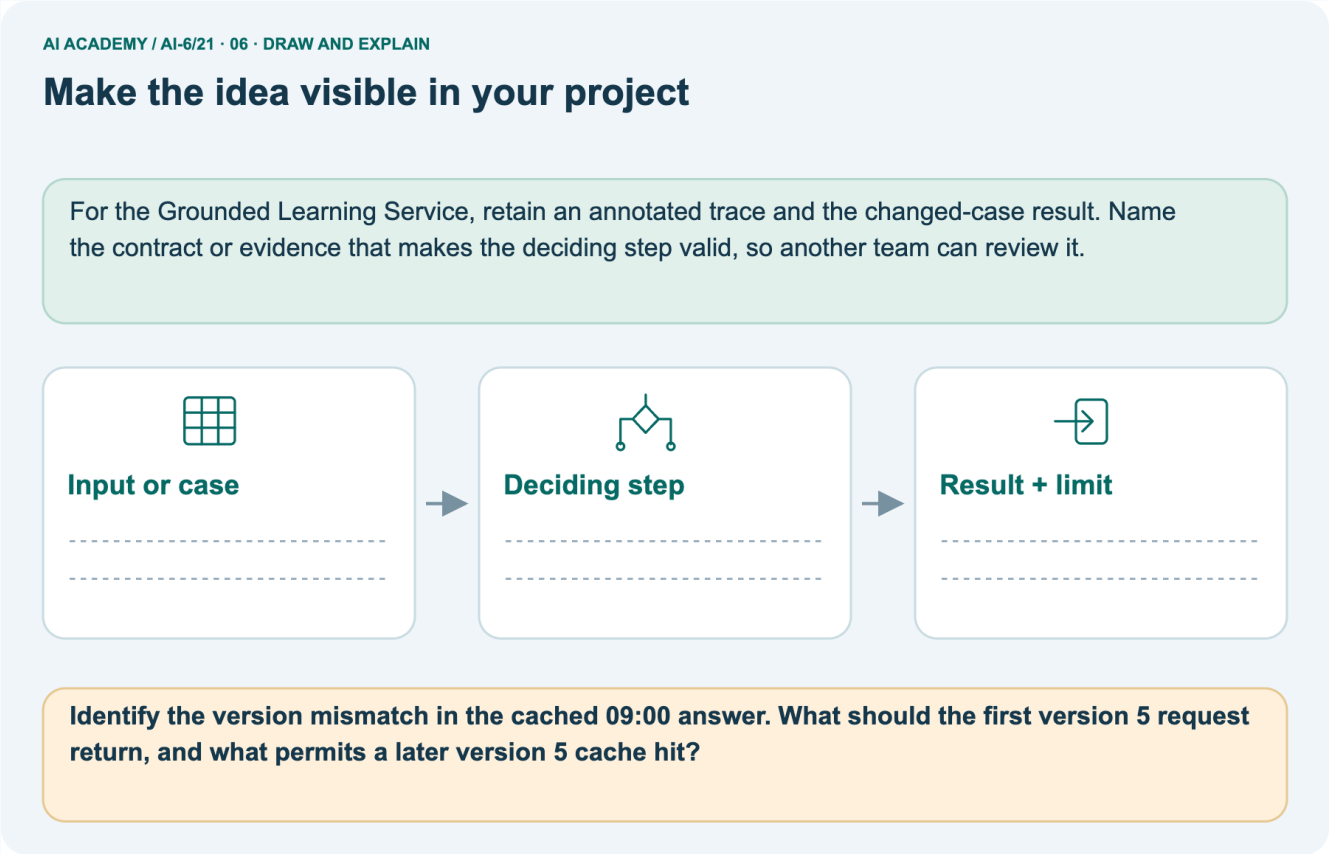
A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

Limit: Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it. Draw your **input**, deciding step, result and limit. Supported question: Identify the version mismatch in the cached 09:00 answer. What should the first version 5 request return, and what permits a later version 5 cache hit?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Caching and stale information: the case

02. Follow the deciding steps

03. Caching and stale information — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.